Akash Fokane

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EDUCATION

Indian Institute of Technology (IIT) Bombay Jul 2019 – Jun 2024
Dual Degree (B.Tech + M.Tech), Electrical Engineering — Specialization: Integrated Circuits *Mumbai*
CGPA: 8 / 10

PROFESSIONAL EXPERIENCE

Hind Rectifiers Jul 2024 – Present
Embedded Engineer *Mumbai*

- Developed bare-metal real-time control firmware on **TI TMS320F28379D** microcontroller (C28x CPU) using **Embedded C**
- Implemented inter-subsystem communication using **CAN**, **MVB** (over SPI), and **Ethernet**, enabling seamless integration and coordination between multiple embedded subsystems
- Configured **Dual-Port RAM (DPRAM)** via the **EMIF** module (**memory-mapped I/O**) for shared memory access between controllers, reducing latency and improving data throughput
- Designed and implemented a custom **Bootloader** for TI TMS320F28379D MCU, supporting firmware updates over **Ethernet**, **CAN**, and **DPRAM**. Developed **linker script modifications** for memory layout control, utilized **TI Flash API** for safe program/erase sequences, and implemented data integrity checks (checksum) to ensure fault-tolerant MCU programming
- Implemented a **data logging** mechanism to record controller state and fault data to external flash for debugging and fault analysis

Samsung Semiconductor India Research May 2022 – Jul 2022
Hardware Engineer *Bangalore*

- Designed and implemented a synchronous **FIFO** buffer in SystemVerilog with proper handshaking, pointer management, and full/empty flag logic; validated corner cases including simultaneous read/write operations and pointer wrap-around
- Developed a constrained **SystemVerilog Testbench** with **Assertions (SVAs)** to verify FIFO functionality and edge conditions, streamlining debugging and ensuring functional correctness through simulation and waveform analysis

PROJECTS

JPEG Encoder on FPGA Jan 2023 – Apr 2023

- Simulated **JPEG** image compression algorithm using Verilog and utilized the **MAX10 FPGA** for hardware implementation
- Implemented 2D **Discrete Cosine Transform** followed by Quantization and RLE encoding for high compression ratio
- Achieved compression ratio of **70%** from a 32×32 pixel image, maintaining its quality without observable degradation

Elevator Controller May 2025

- Implemented a multi-floor elevator controller in **SystemVerilog** using a **FSM** and the **SCAN algorithm** for movement direction, door sequencing, and arbitration of internal/external requests with nearest-floor search
- Developed a **Testbench** in SystemVerilog, incorporating scenarios such as simultaneous up/down requests, boundary floor conditions, and manual door overrides; validated functional correctness using simulated waveform inspection

16-Bit Multiplier & Accumulator Circuit Oct 2022 – Nov 2022

- Designed a 16-bit Multiply and Accumulate Circuit using the **Dadda Algorithm** for optimization of area-time complexity
- Implemented a 16-bit **Brent Kung** logarithmic adder for implementation of the accumulator stage in the MAC Circuit
- Verified the 5-stage MAC circuit using **GHDL** simulation and rigorously tested the design on **50,000+** test cases

Memory Word Game Mar 2021 – Apr 2021

- Developed a word game in **Embedded C** for the **Atmel-AT89C51** microcontroller with I/O via **LCD** module, and programmed it using **Keil-µVision**
- Incorporated **UART** Module and **RealTerm** software for interfacing and testing of the implementation on micro-controller

TECHNICAL SKILLS

Programming Languages	C, Embedded C, Python, MATLAB, Assembly
HDL	Verilog, SystemVerilog, VHDL
MCUs / Processors	TI C2000 (TMS320F28379D), ARM Cortex-M (Tiva C), 8051
FPGA / CPLD	Altera MAX10 (Verilog), Krypton CPLD (VHDL)
Embedded Skills	Bare-metal firmware, ISR-driven design, Memory-Mapped I/O, Bootloader, Flash Programming
Communication Protocols	CAN, SPI, UART, Ethernet
Tools & Debugging	CCS, Keil µVision, ModelSim, Quartus Prime, Ngspice